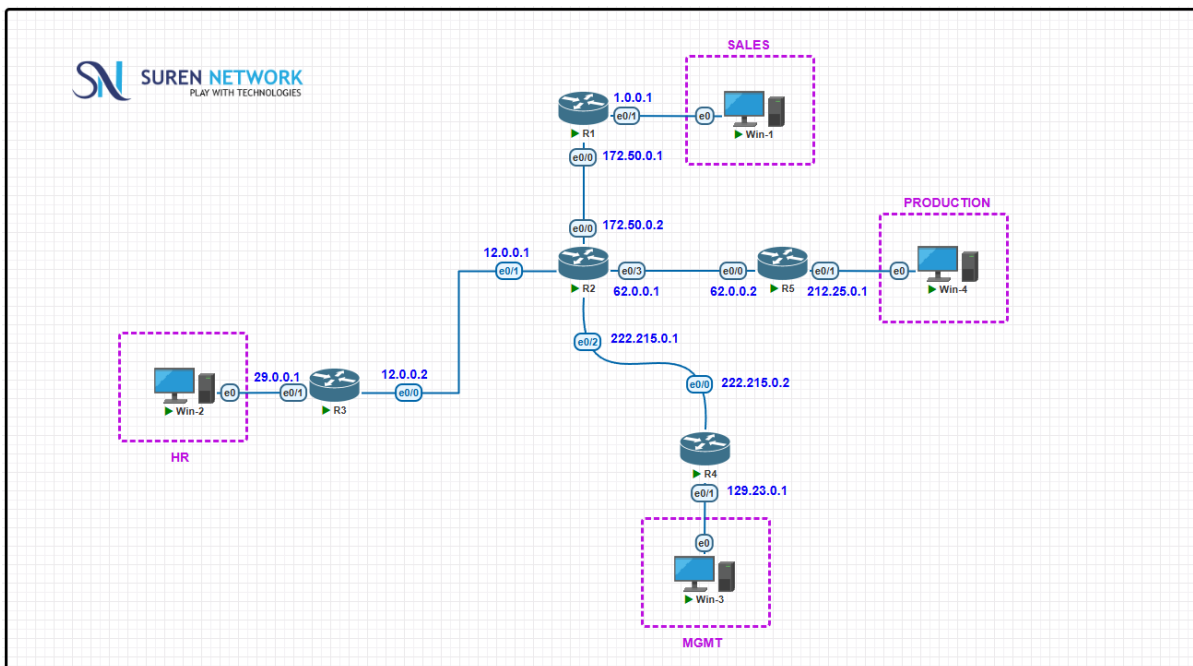


DHCP & RELAY AGENT :



Assign ip address in all Cisco Routers :

R1

R1#

R1#sh ip int br

Interface

ocol

Ethernet0/0

Ethernet0/1

Ethernet0/2

Ethernet0/3

IP-Address

172.50.0.1

1.0.0.1

unassigned

unassigned

OK? Method Status

YES TFTP up

YES TFTP up

YES TFTP down

YES TFTP down

Prot

up

up

down

down

R2

R2#

R2#

R2#sh ip int br

Interface

ocol

Ethernet0/0

Ethernet0/1

Ethernet0/2

Ethernet0/3

IP-Address

172.50.0.2

12.0.0.1

222.215.0.1

62.0.0.1

OK? Method Status

YES TFTP up

YES TFTP up

YES TFTP up

YES TFTP up

Prot

up

up

up

up

R3

R3#sh ip int br

Interface

ocol

Ethernet0/0

Ethernet0/1

Ethernet0/2

Ethernet0/3

IP-Address

12.0.0.2

29.0.0.1

unassigned

unassigned

OK? Method Status

YES TFTP up

YES TFTP up

YES TFTP down

YES TFTP down

Prot

up

up

down

down

R4

R4(config)#do sh ip int br

Interface

ocol

Ethernet0/0

Ethernet0/1

Ethernet0/2

Ethernet0/3

IP-Address

222.215.0.2

129.23.0.1

unassigned

unassigned

OK? Method Status

YES TFTP up

YES TFTP up

YES TFTP down

YES TFTP down

Prot

up

up

down

down

R5

R5#sh

'Sep 14 15:04:27.622: %SYS-5-CONFIG_I: Configured from console by console

R5#sh ip int br

Interface

ocol

Ethernet0/0

Ethernet0/1

Ethernet0/2

Ethernet0/3

IP-Address

62.0.0.2

212.25.0.1

unassigned

unassigned

OK? Method Status

YES TFTP up

YES TFTP up

YES TFTP down

YES TFTP down

Prot

up

up

down

down

Add static route in All routers :

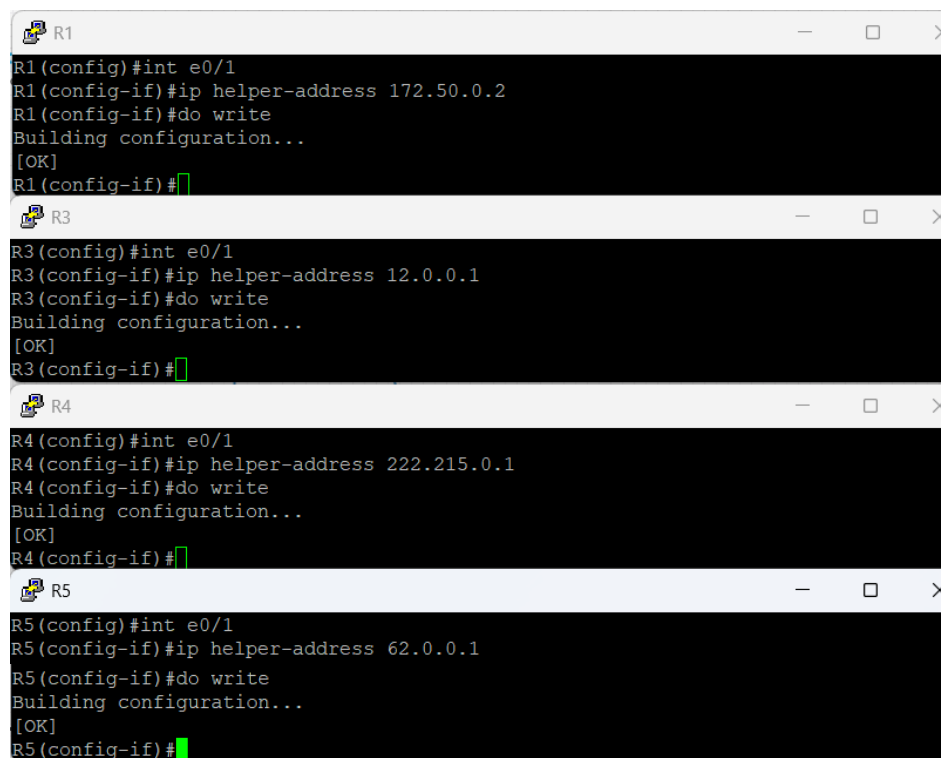
 R1	 R3
Gateway of last resort is not set	Gateway of last resort is not set
S 1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks	S 1.0.0.0/8 [1/0] via 12.0.0.1
C 1.0.0.0/8 is directly connected, Ethernet0/1	12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 1.0.0.1/32 is directly connected, Ethernet0/1	C 12.0.0.0/8 is directly connected, Ethernet0/0
S 12.0.0.0/8 [1/0] via 172.50.0.2	L 12.0.0.2/32 is directly connected, Ethernet0/0
S 29.0.0.0/8 [1/0] via 172.50.0.2	29.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S 62.0.0.0/8 [1/0] via 172.50.0.2	C 29.0.0.0/8 is directly connected, Ethernet0/1
S 129.23.0.0/16 [1/0] via 172.50.0.2	L 29.0.0.1/32 is directly connected, Ethernet0/1
172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks	S 62.0.0.0/8 [1/0] via 12.0.0.1
C 172.50.0.0/16 is directly connected, Ethernet0/0	S 129.23.0.0/16 [1/0] via 12.0.0.1
L 172.50.0.1/32 is directly connected, Ethernet0/0	S 172.50.0.0/16 [1/0] via 12.0.0.1
S 212.25.0.0/24 [1/0] via 172.50.0.2	S 212.25.0.0/24 [1/0] via 12.0.0.1
S 222.215.0.0/24 [1/0] via 172.50.0.2	S 222.215.0.0/24 [1/0] via 12.0.0.1

 R2	 R4	 R5
Gateway of last resort is not set	Gateway of last resort is not set	Gateway of last resort is not set
S 1.0.0.0/8 [1/0] via 172.50.0.1	S 1.0.0.0/8 [1/0] via 222.215.0.1	S 1.0.0.0/8 [1/0] via 62.0.0.1
12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks	S 12.0.0.0/8 [1/0] via 222.215.0.1	S 12.0.0.0/8 [1/0] via 62.0.0.1
C 12.0.0.0/8 is directly connected, Ethernet0/1	S 29.0.0.0/8 [1/0] via 222.215.0.1	S 29.0.0.0/8 [1/0] via 62.0.0.1
L 12.0.0.1/32 is directly connected, Ethernet0/1	S 62.0.0.0/8 [1/0] via 222.215.0.1	62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S 29.0.0.0/8 [1/0] via 12.0.0.2	129.23.0.0/16 is variably subnetted, 2 subnets, 2 masks	C 62.0.0.0/8 is directly connected, Ethernet0/0
62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks	C 129.23.0.0/16 is directly connected, Ethernet0/1	L 62.0.0.2/32 is directly connected, Ethernet0/0
C 62.0.0.0/8 is directly connected, Ethernet0/3	L 129.23.0.1/32 is directly connected, Ethernet0/1	S 129.23.0.0/16 [1/0] via 62.0.0.1
L 62.0.0.1/32 is directly connected, Ethernet0/3	S 172.50.0.0/16 [1/0] via 222.215.0.1	S 172.50.0.0/16 [1/0] via 62.0.0.1
S 129.23.0.0/16 [1/0] via 222.215.0.2	S 212.25.0.0/24 [1/0] via 222.215.0.1	212.25.0.0/24 is variably subnetted, 2 subnets, 2 masks
172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks	222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks	C 212.25.0.0/24 is directly connected, Ethernet0/1
C 172.50.0.0/16 is directly connected, Ethernet0/0	C 222.215.0.0/24 is directly connected, Ethernet0/0	L 212.25.0.1/32 is directly connected, Ethernet0/1
L 172.50.0.2/32 is directly connected, Ethernet0/0	L 222.215.0.2/32 is directly connected, Ethernet0/0	S 222.215.0.0/24 [1/0] via 62.0.0.1
S 212.25.0.0/24 [1/0] via 62.0.0.2		
222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks		
C 222.215.0.0/24 is directly connected, Ethernet0/2		
L 222.215.0.1/32 is directly connected, Ethernet0/2		

Enable dhcp server for all department on R2:

 R2
R2(config)#ip dhcp pool SALES
R2(dhcp-config)#network 1.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 1.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#exit
R2(config)#ip dhcp pool PRODUCTION
R2(dhcp-config)#network 212.25.0.0 255.255.255.0
R2(dhcp-config)#default-router 212.25.0.1
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#exit
R2(config)#
R2(config)#ip dhcp pool MGMT
R2(dhcp-config)#network 129.23.0.0 255.255.0.0
R2(dhcp-config)#default-router 129.23.0.1
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#exit
R2(config)#
R2(config)#
R2(config)#ip dhcp pool HR
R2(dhcp-config)#network 29.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 29.0.0.1
R2(dhcp-config)#lease 1 2 3
R2(dhcp-config)#exit
R2(config)#
R2(config)#do write
Building configuration...
[OK]
R2(config)#

Enable Dhcp Relay Agent on All router :



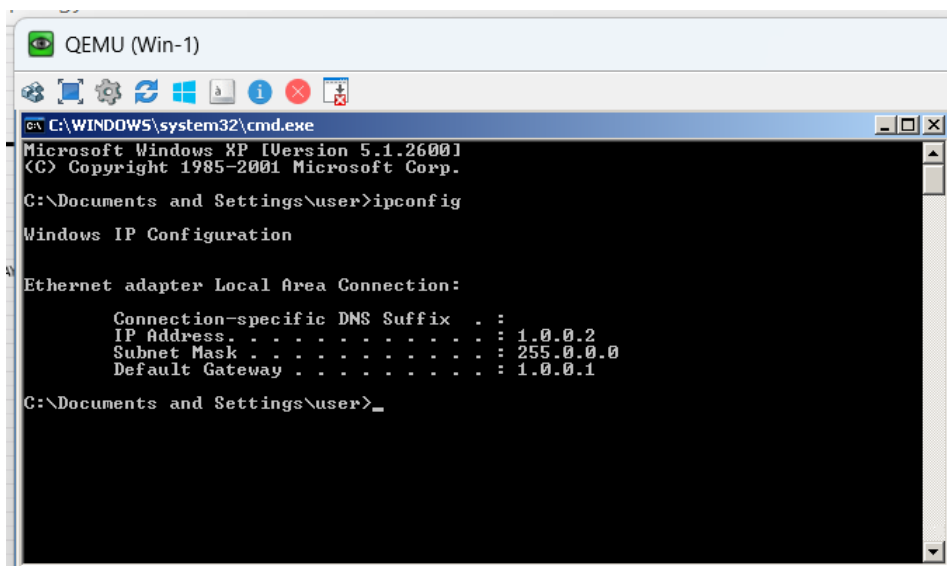
```
R1
R1(config)#int e0/1
R1(config-if)#ip helper-address 172.50.0.2
R1(config-if)#do write
Building configuration...
[OK]
R1(config-if)#

R3
R3(config)#int e0/1
R3(config-if)#ip helper-address 12.0.0.1
R3(config-if)#do write
Building configuration...
[OK]
R3(config-if)#

R4
R4(config)#int e0/1
R4(config-if)#ip helper-address 222.215.0.1
R4(config-if)#do write
Building configuration...
[OK]
R4(config-if)#

R5
R5(config)#int e0/1
R5(config-if)#ip helper-address 62.0.0.1
R5(config-if)#do write
Building configuration...
[OK]
R5(config-if)#
```

Get ip from the dhcp server :



```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 1.0.0.2
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 1.0.0.1

C:\Documents and Settings\user>
```

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 29.0.0.2
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 29.0.0.1

C:\Documents and Settings\user>
```

```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 129.23.0.2
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 129.23.0.1

C:\Documents and Settings\user>
```

Check dhcp datadase in server 2:

```
R2
R2(config)#do sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
1.0.0.2         0150.b7bb.06ac.00 Sep 15 2026 10:20 PM Automatic
29.0.0.2        0150.cd1b.06ad.00 Sep 15 2026 10:20 PM Automatic
129.23.0.2      0150.ce12.06ae.00 Sep 15 2026 10:20 PM Automatic
212.25.0.2      0150.6531.06af.00 Sep 15 2026 10:19 PM Automatic
```